Title 0:00–0:15. Joint work with Ken Brown.

Introduction

0:15–1:00 / 1 Outline Preview the cube, conditions, geometry, completion, and bounds.

1:00–2:00 / 2 Pauli strings X(a), Z(b) commute or anticommute. Stabilizers give the syndrome. Distance concerns nontrivial logical errors.

2:00–3:30 / 3 CSS codes X(a), $a \in C_2$, and Z(b), $b \in C_1^{\perp}$. Logical basis from S and K. Z syndrome Se^{T}.

3:30–4:45 / 4 Transversal gates One gate fault stays on one site. Define CCZ. Eastin–Knill and distillation.

4:45–6:00 / 5 Different codes, different gates Steane: Clifford. 15 qubits: T. Cube: CCZ. State the sign-layer question.

Cube example *6:00–6:10: outline, current theme highlighted.*

6:10–7:15 / 6 Points of $\mathbb{F}_2^3$ Reveal three coordinate faces, then the constant. Their span is RM$(1, 3)$.

7:15–8:00 / 7 Cube encoding S is all ones. K is the coordinates. Each logical state is a complement pair.

8:00–9:00 / 8 Cube distances An edge is a Z logical: d_Z=2. Nonconstant affine codewords give d_X=4.

9:00–10:00 / 9 Signed weights Integer signed weight, mod 8. Every codeword in a logical coset needs the same phase.

10:00–11:30 / 10 Cube signs Put inverse gates on parity p. Distance to p is 4 except for the label 111 pair.

11:30–13:15 / 11 Face, edge, vertex Signed weights: face 0, edge 0, vertex -1. Only the cubic term survives.

13:15–14:00 / 12 Row congruences Coefficients 1, -2, 4 give mod 8, 4, 2. Only the logical triple is odd.

Gate conditions *14:00–14:10: outline, current theme highlighted.*

14:10–14:45 / 13 Historical context Distillation, row constraints, parent spaces, CSS diagonal gates. Keep brief.

14:45–16:00 / 14 Matrix versus space Matrix: distinct rows. Space: all triples, including repeats, so all vectors even.

16:00–17:30 / 15 Nine CH conditions Logical, mixed, stabilizer families. K3 odd. Parities allow a Clifford correction.

Geometric construction *17:30–17:40: outline, current theme highlighted.*

17:40–19:00 / 16 Q48 result 48 physical, 3 logical. d_X=16, d_Z=3. 26 T and 22 inverse, no correction.

19:00–20:00 / 17 Fix the stabilizers Projective columns. 8-divisible span. Find K. Solve signs. Divisibility is a choice.

20:00–21:00 / 18 Projective syndromes Nonzero columns detect one Z. Distinct columns detect two. This proves only d_Z.

21:00–22:00 / 19 48-point support 63 minus 15 gives 48. Remove $w = 00$. Three blocks of 16. Six coordinate rows.

22:00–23:30 / 20 Weights 24 and 32 Nonzero u coefficient: 3 times 8. Pure w coefficient: 2 times 16. Rank six.

23:30–25:00 / 21 Geometric family Recover 48 at $p = 4, q = 2$. Other counts are lower bounds. Star is a fixed-support census.

Column picture and completion *25:00–25:10: outline, current theme highlighted.*

25:10–26:30 / 22 Column indicators Occupied columns as a Boolean function. Cube: y. Q48: $w_1 + w_2 + w_1 w_2$.

26:30–28:00 / 23 Moments and monomials Overlap parity is a monomial moment. One exceptional cubic. State the coset briefly.

28:00–29:30 / 24 Logical labels Three label bits per point. Product forces $x = \kappa$. S-only constraints already hold.

29:30–31:30 / 25 Logical completion One-row space, compatible pairs, linear third row. Three-site witness makes d_Z=3.

31:30–33:00 / 26 $T/T^{\dagger}$ signs Binary gamma in mod-4 equations. Show the polynomial, count $6 + 6 + 10$ inverse gates.

Bounds and questions *33:00–33:10: outline, current theme highlighted.*

33:10–35:00 / 27 Reed–Muller supports Positive-degree tests vanish at 0. Correct weight parity, then use RM duality.

35:00–36:30 / 28 Exclusion through 38 Rank at most 8, support reductions, complete enumeration, no logical completion.

36:30–38:30 / 29 Length comparison Our 38 floor. WJS 39–46. 47 quasi-transversal. Sign failure is code-specific.

38:30–40:00 / 30 Takeaways and questions Geometry, labels, signs. Ask about higher distance and sign feasibility. Leave QR up.

40:00: finish and invite questions.